Lesson 35: Extensive-Form Games and Subgame Perfection

AI/ML Mathematics

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1 Extensive-Form Games

Normal-form games capture simultaneous interaction. Many strategic situations, however, unfold sequentially: players observe previous moves before choosing their own. The extensive form models such sequential structure via game trees.

1.1 Game Trees and Information Sets

Definition 1.1 (Extensive-Form Game)**.** An *extensive-form game* is a tuple $\Gamma = (N, H, P, (\mathcal{I}_i)_{i \in N}, u)$ where:

(i) $N = \{1, \ldots, n\}$ is the set of players. We may include a "chance" player 0 (nature).

(ii) H is a set of *histories* (finite sequences of actions), including the empty sequence $\emptyset$ (the root). H is prefix-closed: if $(a_1, \ldots, a_k) \in H$ then $(a_1, \ldots, a_l) \in H$ for all $l \leq k$. A history $h \in H$ is *terminal* if there is no action a with $(h, a) \in H$. The set of terminal histories is $Z \subseteq H$.

(iii) $P : H \setminus Z \to N \cup \{0\}$ is the *player function*, assigning to each non-terminal history the player who acts there.

(iv) For each $i \in N$, $\mathcal{I}_i$ is a partition of $\{h \in H \setminus Z : P(h) = i\}$ into *information sets*. If h, h' belong to the same information set $I \in \mathcal{I}_i$, then the available actions satisfy $A(h) = A(h')$, denoted $A(I)$.

(v) $u = (u_1, \ldots, u_n) : Z \to \mathbb{R}^n$ assigns payoffs to terminal histories.

Definition 1.2 (Perfect Information)**.** An extensive-form game has *perfect information* if every information set is a singleton: $|I| = 1$ for all $I \in \mathcal{I}_i$, all $i \in N$. At each decision point, the acting player knows the complete history of play.

Definition 1.3 (Perfect Recall)**.** Player i has *perfect recall* if for any two histories h, h' in the same information set $I \in \mathcal{I}_i$, the sequence of player i's information sets and actions along the path from the root to h equals the corresponding sequence along the path to h'. Informally, no player forgets their own past actions or past information.

Example 1.4 (Entry Game)**.** An incumbent (player 2) faces a potential entrant (player 1). The entrant first decides to *Enter* (E) or *Stay Out* (O). If the entrant enters, the incumbent decides to *Fight* (F) or *Accommodate* (A). Payoffs:

- O: $(0, 2)$.
- (E, F): $(-1, -1)$.
- (E, A): $(1, 1)$.

This is a perfect-information game: all information sets are singletons. The histories are $H = \{\emptyset, O, E, (E, F), (E, A)\}$, with $Z = \{O, (E, F), (E, A)\}$.

Example 1.5 (Simultaneous Move as Imperfect Information)**.** A simultaneous 2×2 game can be represented in extensive form as follows: player 1 moves first (choosing T or B), then player 2 moves without observing player 1's choice. Player 2's two decision nodes (after T and after B) belong to a single information set. This shows that *simultaneous* games are a special case of extensive-form games with imperfect information.

1.2 Strategies in Extensive-Form Games

Definition 1.6 (Pure Strategy in Extensive Form)**.** A *pure strategy* for player i in an extensive-form game is a function $s_i : \mathcal{I}_i \to \bigcup_{I \in \mathcal{I}_i} A(I)$ satisfying $s_i(I) \in A(I)$ for each $I \in \mathcal{I}_i$.

Remark 1.7 (Complete Contingent Plans). A strategy specifies an action at *every* information set, including those unreachable given earlier choices. In the Entry Game (Example 1.4), player 2 has two pure strategies (F and A) even though their decision node is reached only if player 1

enters. This apparent redundancy is essential for defining Nash equilibrium and subgame perfection.

Definition 1.8 (Behavioral Strategy)**.** A *behavioral strategy* for player i is a function $b_i : \mathcal{I}_i \to \bigcup_{I \in \mathcal{I}_i} \Delta(A(I))$ satisfying $b_i(I) \in \Delta(A(I))$. That is, b_i assigns a probability distribution over actions at each information set, with independent randomization across information sets.

Proposition 1.9 (Normal-Form Representation)**.** *Every extensive-form game has a corresponding normal-form game: the strategy sets are as in Definition 1.6, and the payoff $u_i(s_1, \ldots, s_n)$ is determined by following the unique path through the tree dictated by the strategy profile (with expectations over chance moves).*

Proof. Given a strategy profile $(s_1, \ldots, s_n)$, define the path from the root inductively. At each non-terminal history h:

- If $P(h) = i \in N$, the next action is $s_i(I)$ where I is the information set containing h.
- If $P(h) = 0$ (chance), the next action is drawn from the prescribed chance distribution.

Since the game tree is finite, this terminates at a terminal history $z \in Z$. Define $u_i(s_1, \ldots, s_n) = \mathbb{E}[u_i(z)]$, where the expectation is over chance moves. The tuple $(N, (S_i), (u_i))$ is a well-defined normal-form game. □

2 Kuhn's Theorem

The relationship between mixed strategies (distributions over pure strategies) and behavioral strategies (local randomization at each information set) is fundamental. In general, behavioral strategies are a strict subset of what mixed strategies can achieve. Kuhn's theorem identifies the exact condition under which they coincide.

Key Result

Theorem 2.1 (Kuhn, 1953)**.** *In a finite extensive-form game with perfect recall, every mixed strategy of player i has a* realization-equivalent *behavioral strategy: a behavioral strategy inducing the same probability distribution over terminal histories for every strategy profile of the opponents. Conversely, every behavioral strategy has a realization-equivalent mixed strategy.*

Proof. We prove both directions.

Direction 1: Mixed to behavioral. Let σ_i be a mixed strategy (a distribution over pure strategies S_i) for player i in a game with perfect recall. We construct a behavioral strategy b_i and show realization equivalence.

For each information set $I \in \mathcal{I}_i$, define the *reach probability* $\rho_i(I) = \sigma_i(\{s_i \in S_i : s_i \text{ reaches } I\})$, where "$s_i$ reaches I" means there exists some opponent profile under which the path of play passes through a history in I when player i follows s_i. Under perfect recall, whether s_i reaches I depends only on s_i's actions at player i's information sets on the path to I (not on actions at information sets after or disjoint from I).

For each action $a \in A(I)$, define:

$$b_i(a|I) = \begin{cases} \dfrac{\sigma_i(\{s_i : s_i(I) = a \text{ and } s_i \text{ reaches } I\})}{\rho_i(I)} & \text{if } \rho_i(I) > 0, \\ \dfrac{1}{|A(I)|} & \text{if } \rho_i(I) = 0. \end{cases}$$

When $\rho_i(I) > 0$, the numerator sums probabilities of pure strategies that both reach I and play a there, so $b_i(\cdot|I)$ is a well-defined distribution.

We verify realization equivalence. The *realization probability* of a terminal history z under σ_i (with opponents playing s_{-i}) is:

$$\mathbb{P}_{\sigma_i}(z \mid s_{-i}) = \sigma_i\big(\{s_i : (s_i, s_{-i}) \text{ leads to } z\}\big).$$

Under the behavioral strategy b_i, the probability of the same terminal history is:

$$\mathbb{P}_{b_i}(z \mid s_{-i}) = \prod_{I \in \mathcal{I}_i^z} b_i(a_I^z \mid I),$$

where $\mathcal{I}_i^z$ is the set of player i's information sets on the path to z, and a_I^z is the action taken at I on this path.

The key step uses perfect recall. Under perfect recall, the information sets $\mathcal{I}_i^z$ are linearly ordered along the path, and the set of pure strategies reaching z factors as:

$$\{s_i : (s_i, s_{-i}) \text{ leads to } z\} = \bigcap_{I \in \mathcal{I}_i^z} \{s_i : s_i(I) = a_I^z \text{ and } s_i \text{ reaches } I\}.$$

By perfect recall, knowing that s_i reaches I and plays a_I^z there determines that s_i reaches the next information set I' on the path. This gives a telescoping product:

$$\begin{aligned}
\mathbb{P}_{\sigma_i}(z \mid s_{-i}) &= \sigma_i\Bigg(\bigcap_{I \in \mathcal{I}_i^z} \{s_i : s_i(I) = a_I^z, s_i \text{ reaches } I\}\Bigg) \\
&= \prod_{I \in \mathcal{I}_i^z} \frac{\sigma_i(\{s_i : s_i(I) = a_I^z, s_i \text{ reaches } I\})}{\sigma_i(\{s_i : s_i \text{ reaches } I\})} \\
&= \prod_{I \in \mathcal{I}_i^z} b_i(a_I^z \mid I) = \mathbb{P}_{b_i}(z \mid s_{-i}).
\end{aligned}$$

The second equality uses the chain rule and the perfect recall structure: the conditional probability of playing a_I^z at I given reaching I is independent of what happens at other information sets not on the path between the previous information set and I.

Direction 2: Behavioral to mixed. Given a behavioral strategy b_i, define a mixed strategy by $\sigma_i(s_i) = \prod_{I \in \mathcal{I}_i} b_i(s_i(I)|I)$. This treats each information set as an independent randomization device. By construction, the probability that the path passes through any sequence of information-set/action pairs is the same product under both σ_i and b_i, establishing realization equivalence. □

Remark 2.2 (Computational Significance). A behavioral strategy requires $O(\sum_I |A(I)|)$ parameters (one distribution per information set), while a mixed strategy over pure strategies may require exponentially many parameters (one probability per complete contingent plan). Kuhn's theorem says we lose nothing by working with the compact behavioral representation.

Proposition 2.3 (Failure Without Perfect Recall)**.** *In games without perfect recall, mixed and behavioral strategies may yield different sets of reachable outcome distributions.*

Proof. Consider a one-player game where the player moves twice but forgets their first action. Specifically: the player first chooses L or R, then (in a single information set containing both post-L and post-R nodes) chooses A or B. Payoffs: $(L, A) \mapsto 1$, $(L, B) \mapsto 0$, $(R, A) \mapsto 0$, $(R, B) \mapsto 1$.

A behavioral strategy specifies $p = \mathbb{P}(L)$ at the first node and $q = \mathbb{P}(A)$ at the second information set. The expected payoff is $pq + (1-p)(1-q) = 1 - p - q + 2pq$, which has maximum $1/2$ (at $p = q = 1/2$).

The mixed strategy playing (L, A) with probability $1/2$ and (R, B) with probability $1/2$ achieves expected payoff $1/2 \cdot 1 + 1/2 \cdot 1 = 1$. This correlates the two moves, which no behavioral strategy can do, since behavioral strategies randomize independently at each information set. □

RL/ML Connection

A behavioral strategy is precisely an RL policy: it maps information states (information sets) to distributions over actions. Kuhn's theorem justifies the standard RL practice of parameterizing policies as functions $\pi(a|s)$ rather than distributions over complete plans. The perfect recall condition corresponds to the Markov property: the agent's policy depends only on the current information state, not on the randomization device used to reach it.

3 Subgame Perfect Equilibrium

3.1 Non-Credible Threats

Example 3.1 (Non-Credible Threat in the Entry Game)**.** In the Entry Game (Example 1.4), consider the strategy profile (O, F): player 1 stays out, player 2 threatens to fight. Payoffs: $(0, 2)$.

Is this a NE? Player 1's deviation to E yields -1 (since player 2 plays F), so player 1 has no incentive to deviate. Player 2's information set is never reached, so any strategy gives the same outcome. Hence (O, F) is a NE. However, if player 1 did enter, player 2 would prefer to accommodate (payoff 1) over fighting (payoff -1). The threat to fight is *non-credible*: it would not be carried out if the relevant decision point were actually reached. Subgame perfection eliminates such equilibria.

3.2 Subgames

Definition 3.2 (Subgame)**.** A *subgame* of Γ beginning at history $h \in H \setminus Z$ is the restriction of Γ to histories extending h, provided:

(i) $\{h\}$ is a singleton information set for player $P(h)$.

(ii) Every information set containing a successor of h is entirely contained within the subtree rooted at h.

The subgame inherits the player function, information partition (restricted to the subtree), and payoffs from Γ.

Remark 3.3. In a game with perfect information, every non-terminal history begins a subgame (since all information sets are singletons and condition (ii) is automatic). In games with imperfect information, subgames may be rare. The entire game (rooted at $\emptyset$) is always a subgame.

Definition 3.4 (Subgame Perfect Equilibrium)**.** A strategy profile s^* is a *subgame perfect equilibrium* (SPE) if its restriction to every subgame is a Nash equilibrium of that subgame.

Proposition 3.5 (SPE Refines NE)**.** *Every subgame perfect equilibrium is a Nash equilibrium.*

Proof. The entire game is a subgame (beginning at the root $\emptyset$). If s^* is an SPE, then by definition it is a NE of every subgame, including the entire game. □

Example 3.6 (SPE in the Entry Game)**.** The Entry Game has two subgames: the whole game (rooted at $\emptyset$) and the subgame starting after E (where player 2 chooses F or A).

In the subgame after E: player 2 chooses A (payoff $1 > -1$). The unique NE of this subgame is A.

In the full game: player 1 anticipates player 2 accommodating, so entry gives payoff $1 > 0$ (staying out). Hence player 1 enters.

The unique SPE is (E, A) with payoffs $(1, 1)$. The NE (O, F) is eliminated because F is not a NE in the post-entry subgame.

3.3 Backward Induction

Definition 3.7 (Backward Induction)**.** For a finite game with perfect information, *backward induction* proceeds as follows:

1. Identify all *penultimate nodes*: non-terminal histories h such that all successors (h, a) are terminal. At each such h, player $P(h)$ chooses the action maximizing their payoff: $a^*(h) = \arg\max_{a \in A(h)} u_{P(h)}(h, a)$.
2. Replace each penultimate node h with a terminal node having payoff $u(h, a^*(h))$.
3. Repeat until the root is reached.

The resulting strategy profile assigns an action at every node of the original tree.

Key Result

Theorem 3.8 (Backward Induction Yields SPE)**.** *In a finite game with perfect information and no ties (the payoff function is injective when restricted to any player's choices at any node), backward induction produces the unique subgame perfect equilibrium.*

Proof. We proceed by strong induction on the depth d of the game tree (the maximum number of actions in any history).

Base case ($d = 1$): The game consists of a single player choosing at the root, with all successors terminal. Backward induction selects the payoff-maximizing action, which is unique (no ties). This is trivially the unique SPE.

Inductive step: Assume the result holds for all games of depth $< d$. Consider a game Γ of depth d with perfect information and no ties. At the root, player $P(\emptyset)$ has actions $A(\emptyset) = \{a_1, \ldots, a_l\}$. Each action a_j leads to a subgame Γ_{a_j} of depth $< d$. By the inductive hypothesis, backward induction produces the unique SPE $s^*_{a_j}$ of Γ_{a_j}, with payoff vector $v(a_j) \in \mathbb{R}^n$.

Backward induction assigns to the root the action $a^* = \arg\max_{a_j} v_{P(\emptyset)}(a_j)$, which is unique since there are no ties. Define s^* as: play a^* at the root, and follow $s^*_{a_j}$ in each subgame Γ_{a_j}.

s^* is an SPE: In each subgame Γ_{a_j}, the restriction of s^* is $s^*_{a_j}$, an SPE by hypothesis. In the full game: the root player chooses a^* optimally given the unique SPE continuations, and no other player moves at the root. Hence no player has a profitable deviation in any subgame.

Uniqueness: Any SPE $\hat{s}$ must restrict to the unique SPE in each Γ_{a_j} (by the inductive hypothesis), so the root player faces the same payoffs $v(a_j)$ and, with no ties, must choose a^*. Hence $\hat{s} = s^*$. □

4 Zermelo's Theorem

Key Result

Theorem 4.1 (Zermelo, 1913)**.** *Every finite extensive-form game with perfect information has a pure strategy subgame perfect equilibrium. If the game also has no ties (each player has distinct payoffs at all terminal nodes reachable from any decision node), the SPE is unique.*

Proof. **Existence.** We prove by strong induction on $|H \setminus Z|$, the number of non-terminal histories.

Base case ($|H \setminus Z| = 1$): The game has a single decision node (the root) with terminal successors. The acting player has at least one payoff-maximizing action (finite, non-empty action set). Any such action constitutes a pure strategy SPE.

Inductive step: Let Γ have $|H \setminus Z| = k > 1$. The root player $P(\emptyset)$ has actions $a_1, \ldots, a_l$. Each a_j leads to a subgame Γ_{a_j} with strictly fewer non-terminal histories. By the inductive hypothesis, each Γ_{a_j} has a pure strategy SPE $s^*_{a_j}$ with payoff vector $v(a_j)$.

Define s^*: at the root, choose $a^* \in \arg\max_{a_j} v_{P(\emptyset)}(a_j)$ (non-empty since l is finite); in each subgame Γ_{a_j}, follow $s^*_{a_j}$.

We verify s^* is an SPE. For any subgame not containing the root, the restriction of s^* is $s^*_{a_j}$ for the appropriate j, which is an SPE by hypothesis. For the entire game: the root player optimizes given the SPE continuations (by construction), and no other player moves at the root. Hence s^* is an SPE.

Uniqueness under no ties. By Theorem 3.8, backward induction produces the unique SPE when all choices have distinct payoffs. □

Corollary 4.2 (Chess Has a Determined Outcome)**.** *In chess (viewed as a finite game with perfect information—the 50-move rule and threefold repetition ensure finiteness), exactly one of the following holds: White has a winning strategy, Black has a winning strategy, or both players can force a draw.*

Proof. Chess is a two-player, finite, perfect-information game with three possible outcomes: White wins, Black wins, or draw. By Zermelo's theorem, a pure strategy SPE exists. In this SPE, the outcome is determined (it is one of the three possibilities). The SPE strategy for the favored side (or both sides in the draw case) guarantees this outcome regardless of the opponent's play, since in any subgame the SPE prescribes optimal behavior. □

Remark 4.3 (Zermelo vs. Minimax). For two-player zero-sum perfect-information games, Zermelo's theorem and the minimax theorem (Lesson 34, Theorem **??**) are related but distinct. Zermelo guarantees a pure SPE in the extensive form. The minimax theorem guarantees saddle-point equality in the normal form (possibly requiring mixed strategies in general). For finite perfect-information zero-sum games, both results apply, and the backward induction value equals the minimax value.

Example 4.4 (Tic-Tac-Toe)**.** Tic-tac-toe is a finite, two-player, perfect-information game with three outcomes (X wins, O wins, draw). By Zermelo's theorem, the game has a determined outcome. It is known (by exhaustive computation of the game tree, which has fewer than $9! = 362880$ terminal nodes) that both players can force a draw. The SPE consists of optimal strategies guaranteeing at least a draw for each player.

5 Connections to Reinforcement Learning

RL/ML Connection

MDPs as single-player extensive-form games. A finite-horizon MDP with deterministic transitions is a single-player game with perfect information. The "player" is the agent, the "histories" are sequences of state-action pairs, and the "payoff" at a terminal history is the cumulative reward. Backward induction is precisely dynamic programming: the Bellman equation $V_t(s) = \max_a[r(s,a) + V_{t+1}(s')]$ computes the optimal action at each state by working backward from the horizon.

With stochastic transitions, nature (player 0) acts between the agent's moves, introducing chance nodes. The backward induction computation becomes $V_t(s) = \max_a[r(s,a) + \mathbb{E}_{s'}[V_{t+1}(s')]]$, which is the standard Bellman optimality equation.

RL/ML Connection

Multi-agent backward induction. In multi-agent settings (stochastic games), backward induction at each state requires solving a game rather than an optimization. This is the foundation of:

- **Minimax-Q** [Shoham and Leyton-Brown, 2008]: backward induction with minimax stage games in zero-sum stochastic games.
- **AlphaGo/AlphaZero**: Monte Carlo Tree Search (MCTS) approximately performs backward induction on enormous game trees by sampling trajectories and using neural network value estimates.
- **Subgame solving** in poker: computing SPE (or approximations) in subgames of imperfect-information games.

6 Exercises

Exercise 6.1 (Normal-Form Representation)**.** Consider the Entry Game (Example 1.4).

(a) List all pure strategies for each player.
(b) Write the normal-form (bimatrix) representation.
(c) Find all pure strategy NE of the normal form.
(d) Identify which NE are subgame perfect.

Exercise 6.2 (Three-Stage Backward Induction)**.** Player 1 chooses A or B. If A: player 2 chooses C or D. If B: player 2 chooses E or F. If $A \to D$: player 1 chooses G or H. Terminal payoffs (player 1, player 2):

- $A \to C$: $(3, 2)$, $A \to D \to G$: $(4, 1)$, $A \to D \to H$: $(1, 3)$.
- $B \to E$: $(2, 4)$, $B \to F$: $(5, 0)$.

(a) Apply backward induction step by step.
(b) Write the complete SPE strategy for each player.
(c) How many pure strategy NE does the normal form have? Which are subgame perfect?

Exercise 6.3 (Kuhn's Theorem Verification)**.** In a two-player game, player 1 observes a private type $t \in \{H, L\}$ (each equally likely) and sends a message $m \in \{h, l\}$. Player 2 observes the message but not the type, and chooses U or D.

(a) How many pure strategies does each player have?
(b) Verify that this game has perfect recall.
(c) Construct a behavioral strategy for player 1 that plays h with probability $3/4$ when $t = H$ and l with probability 1 when $t = L$. Write the realization-equivalent mixed strategy over pure strategies.
(d) Explain why Kuhn's theorem applies and state its consequence.

Exercise 6.4 (Non-Credible Threats in a Richer Game)**.** An entrant chooses E or O. If E, the incumbent chooses F (fight), A (accommodate), or P (price war). Payoffs:

- O: $(0, 3)$, (E, F): $(-2, -1)$, (E, A): $(1, 1)$, (E, P): $(-3, -2)$.

(a) Find all pure strategy NE of the normal-form representation.
(b) Identify which NE involve non-credible threats and explain why.
(c) Find the unique SPE.

Exercise 6.5 (Subgame Count)**.** Consider a three-player game: player 1 chooses L or R. If L: players 2 and 3 move simultaneously (player 2 chooses A or B, player 3 chooses X or Y, neither observing the other's choice). If R: player 2 chooses C or D (with player 3 not moving).

(a) Draw the game tree with information sets.
(b) How many subgames does this game have?
(c) Explain why the simultaneous move after L does not generate additional subgames.

Exercise 6.6 (Zermelo and Nim)**.** In the game of Nim with a single pile of n stones, two players alternate removing 1, 2, or 3 stones. The player who takes the last stone wins (payoff 1; the loser gets 0).

(a) Show this is a finite perfect-information game.
(b) Use backward induction to determine the winner for $n = 1, 2, 3, 4$.
(c) Prove by induction that player 1 wins if and only if n is not divisible by 4.
(d) Relate this to Zermelo's theorem.

Exercise 6.7 (SPE in a Bargaining Game)**.** Two players divide a dollar. In round 1, player 1 proposes a split $(x, 1-x)$ with $x \in \{0, 0.5, 1\}$. Player 2 accepts (game ends with these payoffs) or rejects. If rejected, in round 2, player 2 proposes $(y, 1-y)$ with $y \in \{0, 0.5, 1\}$, and player 1 accepts or rejects. If rejected again, both get 0.
(a) Model this as an extensive-form game and count the subgames.
(b) Apply backward induction to find the SPE.
(c) How does the SPE change if round 2 payoffs are discounted by $\delta \in (0, 1)$?

Exercise 6.8 (Failure of SPE Refinement Under Imperfect Information)**.** Construct a two-player extensive-form game with imperfect information that has exactly one subgame (the entire game) and a NE in which some player's strategy is not "sequentially rational" at a non-singleton information set. Explain why SPE fails to refine NE in this case and what stronger solution concept is needed.

Summary. This lesson developed: (1) the extensive-form game model with histories, information sets, and the distinction between perfect and imperfect information; (2) pure and behavioral strategies, with Kuhn's theorem establishing their equivalence under perfect recall; (3) subgame perfection as a refinement of Nash equilibrium that eliminates non-credible threats; (4) backward induction as a constructive algorithm that produces the unique SPE in perfect-information games without ties; and (5) Zermelo's theorem guaranteeing SPE existence in all finite perfect-information games. The connection to RL is direct: backward induction is single-agent dynamic programming, and behavioral strategies are RL policies.

References

Yoav Shoham and Kevin Leyton-Brown. *Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations.* Cambridge University Press, 2008.